A SEMINAR REPORT ON

RASPBERRY PI BASED WEATHER REPORTING OVER IOT

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE
IN THE PARTIAL FULFILLMENT FOR THE AWARD OF THE DEGREE

OF

FINAL YEAR
IN
E & TC ENGINEERING

By

MR.KASHID VISHAL MOHAN
MR.KATE SAGAR HARIDAS
MR.LALAGE TUSHAR DASHARATH

UNDER THE GUIDANCE OF
DR.N.B.DHAIGUDE

DEPARTMENT OF E & TC ENGINEERING
SVPM'S COLLEGE OF ENGINEERING
MALEGAON(BK)
A/P-MALEGAON(BK)-413115, TAL. BARAMATI, DIST. PUNE (MS) INDIA

2024-25

CERTIFICATE

This is to certify that the Seminar report entitled

RASPBERRY PI BASED WEATHER REPORTING OVER IOT

Submitted By

Kashid Vishal Mohan (B400190140)
Kate Sagar Haridas (B400190141)
Lalage Tushar Dasharath (B400190148)

is a bonafide work carried out by them under the supervision of Prof. Name of Guide and it is approved for the partial fulfillment of the requirement of Savitribai Phule Pune University.

This Seminar report has not been earlier submitted to any other Institute or University for the award of Completed Project Based Learning Work.

Dr.N.B Dhaigude	**Dr.R.K.Patil**
Internal Guide	Head of Department
Department of E & TC	Department of E & TC
....	**Prof. Dr.Mukane S.M.**
External Examiner	Principal
Date:	SVPM's College of Engg. Malegaon(BK)

Place: SVPM's College of Engineering,Malegaon(Bk).
Date:

Acknowledgements

We take this opportunity to thank our project guide Prof.Patil R. K. and Head of the Department Prof.Jagtap K.G. for their valuable guidance and for providing all the necessary facilities, which were indispensable in the completion of this project report. We are also thankful to all the staff members of the Department of E & TC of SVPM's College of Engineering, Malegaon(Bk) for their valuable time, support, comments, suggestions and persuasion. We would also like to thank the institute for providing the required facilities, Internet access and important books.

Kashid Vishal Mohan
Kate Sagar Haridas
Lalage Tushar Dasharath

Abstract

In this proposed is an advanced solution for monitoring the weather conditions at a particular place and make the information visible anywhere in the world. The technology behind this is Internet of Things (IoT), which is an advanced and efficient solution for connecting the things to the internet and to connect the entire world of things in a network Here things might be whatever like electronic gadgets, sensors and automotive electronic equipment. The system deals with monitoring and controlling the environmental conditions like temperature, relative humidity, light intensity and CO level with sensors and sends the information to the web page and then plot the sensor data as graphical statistics. The data updated from the implemented system can be accessible in the internet from anywhere in the world. Weather condition plays an very important role in our daily life. Collecting of data about the different parameters of the weather is necessary for planning in home and environments. Recent developments in Internet of Things made possible to collect the data. In this system some digital as well as analogue sensors like DHT11, BMP180, LDR and marked scale with ULN2803 are used for environmental parameter measuring. This data from input sensors is then read by server that is Raspberry Pi itself and stored in CSV as well as text files. The sensors gather the data of various environmental parameters and provide it to Raspberry PI which act as a base station. The Raspberry PI then transmits the data using WIFI and the processed data will be displayed on laptop through accessing the server that is on the receiver side.

Contents

1 Preamble **1**
1.1 Introduction 1
1.2 Literature Survey 2
1.3 Problem Statement 3
1.4 Objectives 3
1.4.1 Hypothesis 4

2 Block Diagram and Its Explanation **6**
2.1 Block Diagram 6

3 Component Used **9**

4 Circuit Diagram **16**
4.1 Circuit Diagram 16

5 Algorithm **20**

6 Flowchart **21**

7 Result **23**

8 Advantages, Disadvantages and Applications **26**
8.1 Advantages 26
8.2 Disadvantages 26
8.3 Applications 27

9 Future Scope **28**

10 Conclusion **30**

11 References **31**

12 Publication Details **33**

List of Symbols, Abbreviations and Nomenclature

Symbol	**Details**
IoT	Internet of Things
IDE	Integrated Development Environment
GPIO	General Purpose Input Output
ADC	Analog to Digital Converter
HDMI	High Definition Multimedia Interface
USB	Universal Serial Bus
LCD	Liquid Crystal Display

List of Figures

2.1 Block Diagram 6

3.1 Raspberry Pi 4 Model B 9
3.2 DHT11 Sensor 10
3.3 Raindrop Sensor 11
3.4 16*2 LCD Display 12
3.5 I2C Module 13
3.6 Power Supply 14

4.1 Circuit Diagram 16

6.1 Flowchart 21

7.1 Demonstration 23
7.2 Output of the ThingSpeak IoT 25

12.1 Certificate 1 34
12.2 Certificate 2 34
12.3 Certificate 3 35

Chapter 1

Preamble

1.1 Introduction

Weather or Climate is important part of human life.Sensors are essential components not only applicable to the industries for process control but also in daily life for safety of building's and traffic flow measuring, environmental parameters measurement. IoT means Internet of Things. It provides internetworking of physical devices, buildings, vehicles and other components like sensors and actuators.By giving network connectivity to systems embedded with electronics, software, sensors and actuators; these objects are able to collect and exchange data. By using IoT objects to be sensed or controlled remotely through existing network. It gives opportunity to connect physical world with computer-based systems.With the increasing impact of climate change and unpredictable weather patterns, real-time weather monitoring and forecasting have become more critical than ever. Traditional weather stations are often expensive, require significant infrastructure, and are limited in deployment due to size and cost constraints. There is a growing need for compact, cost-effective, and scalable solutions to monitor environmental conditions, especially in remote or underdeveloped areas.

The advancement of Internet of Things (IoT) technologies has opened up new possibilities for remote sensing and data collection. IoT enables everyday objects to be connected to the internet, allowing them to send and receive data without human intervention. When integrated with environmental sensors, IoT devices can be used to monitor parameters like temperature, humidity, atmospheric pressure, and more.

Raspberry Pi, a low-cost, credit-card-sized computer, serves as an excellent platform for developing IoT-based weather monitoring systems. Its GPIO (General Purpose Input/Output) pins allow easy interfacing with various sensors, and its built-in networking capabilities (Wi-Fi or Ethernet) make it ideal for uploading data to the cloud or an online dashboard for remote access.

This project leverages the Raspberry Pi to collect real-time weather data using sensors like DHT11/DHT22 (for temperature and humidity), BMP180/BMP280 (for atmospheric pressure), and potentially other sensors such as a rain gauge or wind speed sensor. The data collected is then transmitted over the internet and displayed on a web-based interface or IoT platforms like ThingSpeak, Blynk, or Firebase. Users can view the weather information from anywhere in the world via their smartphones or computers.

This kind of system finds applications in:

1. Smart agriculture
2. Remote weather monitoring
3. Disaster management
4. Educational and research purposes
5. Smart city infrastructure

1.2 Literature Survey

1.paper Name : IOT Based Weather Monitoring and Reporting System Project
Author Name : Bhagat AM, Thakare AG, KAM, Muneshwar NS, and Choudhary PV
Description : It synthesizes key findings from various studies, highlighting methodologies, results, and emerging trends. The authors identify critical gaps in the existing research, emphasizing the need for further exploration in areas.
2.paper Name : Prolong Network Lifetime Using Dynamic Routing for Wireless Sensor Networks
Author Name : U. Hariharan and K. Rajkumar

Description : The authors identify gaps in the current literature, underscoring areas that require further investigation. By providing a comprehensive overview, this survey serves as a foundational resource for researchers and practitioners, facilitating informed discussions and guiding future research efforts.
3.paper Name : IoT Based Weather Monitoring System for Effective Analytics
Author Name : Joseph FJJ
Description : The survey identifies significant gaps in the existing research, suggesting areas that warrant further exploration.
4.paper Name : Iot Based Weather Station Using Raspberry Pi 3
Author Name : Muck PY and Homam MJ's
Description : This survey serves as an important resource for researchers and practitioners, facilitating a deeper understanding and guiding future investigations in the field.
5.paper Name : Multimodal Biometric Authentication System using Deep Learning Method
Author Name : S. S. Sengar, U. Hariharan, and K. Rajkumar
Description : By providing a thorough overview of existing studies, this survey serves as a valuable reference for researchers and practitioners, fostering a deeper understanding and guiding subsequent research efforts in the area.

1.3 Problem Statement

Existing weather reporting systems primarily rely on centralized meteorological stations that may not provide localized data. This can result in a lack of awareness about immediate weather changes, which can adversely affect farmers, outdoor workers, and communities vulnerable to severe weather events.

1.4 Objectives

The primary objective of this project is to design and implement a smart, real-time weather monitoring system using a Raspberry Pi and Internet of Things (IoT) technology. The system aims to measure essential environmental parameters such as temperature, humidity, atmospheric pressure, and air quality using various sensors connected to the Raspberry Pi.

One of the key objectives is to collect accurate weather data and process it locally using the Raspberry Pi's computing capability. The system should be able to transmit this data over the internet to a cloud platform for storage, analysis, and visualization. The goal is to ensure that the data is accessible from anywhere in real time via web browsers or mobile applications.

Another important objective is to create a user-friendly interface (dashboard or app) for live monitoring of weather conditions. The project also aims to set up threshold values for each parameter and generate alerts or notifications when those values are exceeded, helping users take timely actions in sensitive applications such as agriculture or disaster prevention.

Furthermore, the system should be scalable to include more sensors or additional Raspberry Pi nodes in the future. Power efficiency and reliability are also key objectives, particularly for deployment in remote or rural locations. The project also seeks to maintain data logs for historical weather analysis, which could be used for trend prediction or climate studies.

Additionally, the objective is to use open-source tools and platforms to reduce cost and ensure flexibility in development. By implementing this system, the project intends to demonstrate how IoT can be effectively used in environmental monitoring and how embedded systems like Raspberry Pi can replace expensive traditional weather stations.

In summary, the project aims to provide a low-cost, real-time, remotely accessible, and reliable weather reporting solution that can be used in various domains such as smart agriculture, urban planning, environmental research, and early warning systems.

1.4.1 Hypothesis

The hypothesis of this project is that a low-cost, compact, and energy-efficient system for real-time weather monitoring can be developed using a Raspberry Pi integrated with IoT technology. It is assumed that by interfacing various environmental sensors—such as temperature, humidity, atmospheric pressure, and gas sensors—with the Raspberry Pi, accurate weather data can be collected continuously. This data can then be processed and transmitted wirelessly over the internet using the Raspberry Pi's built-in Wi-Fi or Ethernet capabilities.

It is further hypothesized that cloud platforms like ThingSpeak, Blynk, or Firebase can be used effectively to store, visualize, and analyze weather data remotely. The system will enable users to access live weather updates from any location through web browsers or mobile applications. Real-time notifications and alerts can also be implemented if weather parameters exceed predefined thresholds. This will enhance the usefulness of the system for applications like agriculture, disaster management, and smart city planning.

Moreover, the project assumes that the Raspberry Pi can act as both a data logger and an IoT gateway, capable of handling sensor input, data processing, and cloud communication simultaneously. It is also expected that the system will be reliable, scalable, and customizable, allowing additional sensors or features to be integrated easily. With proper error handling and power management, the system could be deployed in remote areas with minimal maintenance. Thus, the overall hypothesis is that an IoT-enabled weather monitoring system based on Raspberry Pi can serve as a cost-effective, real-time, and accessible solution for environmental data reporting.

Chapter 2

Block Diagram and Its Explanation

2.1 Block Diagram

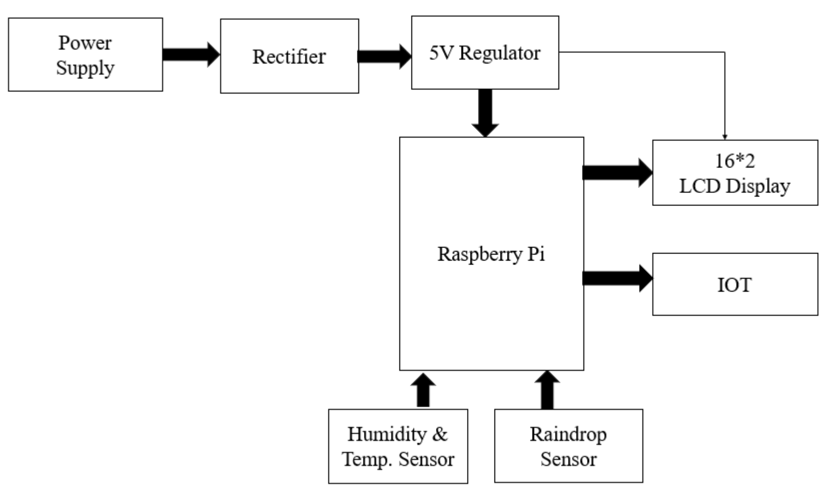

Figure 2.1: Block Diagram

1.Power Supply

Type : Electrical source (e.g., AC mains)

Dimensions : Varies, typically 230V AC input

Function : Provides raw power input to the circuit

Connection : Connected to the Rectifier block

Key Point : It is the starting point; supplies energy to the whole system.

2.Rectifier

Type : AC to DC Converter (Bridge Rectifier)

Dimensions : Depends on component (e.g., 1A–2A current rating)

Function : Converts AC power to unregulated DC

Connection : Input from Power Supply, Output to 5V Regulator

Key Point : Essential for powering DC electronic components.

3.5V Regulator

Type : Voltage Regulator (e.g., 7805 IC)

Dimensions : Standard TO-220 package

Function : Regulates and supplies constant 5V DC

Connection : Input from Rectifier, Output to Raspberry Pi and LCD

Key Point : Maintains stable voltage for safe Raspberry Pi operation.

4.Raspberry Pi

Type : Mini Computer/Microcontroller (e.g., Raspberry Pi 3)

Dimensions : $\tilde{8}5$mm × 56mm

Function : Controls entire system, processes sensor data, sends to IoT

Connection : Input from sensors, Output to LCD IoT, Powered by 5V Regulator

Key Point : Central control unit of the project.

5.Humidity Temperature Sensor

Type : Environmental Sensor (e.g., DHT11/DHT22)

Dimensions : Small module ($\tilde{1}5$mm × 12mm)

Function : Measures ambient temperature and humidity

Connection : Connected to GPIO pins of Raspberry Pi

Key Point : Provides key weather parameters.

6.Raindrop Sensor

Type : Water Detection Sensor

Dimensions : $\tilde{4}0$mm $\times$ 20mm module

Function : Detects presence and intensity of rain

Connection : Connected to Raspberry Pi GPIO

Key Point : Helps detect rainfall in real-time.

7.16×2 LCD Display

Type : Display Module (HD44780 Controller)

Dimensions : $\tilde{8}0$mm $\times$ 36mm

Function : Displays live weather data (temp, humidity, rain)

Connection : Connected to Raspberry Pi and powered by 5V Regulator

Key Point : Provides local, visual output of sensor data.

8.IoT Module

Type : Wi-Fi Module (e.g., onboard or external like ESP8266)

Dimensions : $\tilde{2}5$mm $\times$ 15mm (if external)

Function : Sends weather data to the internet/cloud

Connection : Connected to Raspberry Pi via GPIO or USB

Key Point : Enables remote access and cloud storage of data.

Chapter 3

Component Used

1.Raspberry Pi 4 Model B

2.DHT11 Sensor

3.Raindrop Sensor

4.16*2 LCD Display

5.I2C

6.ThingSpeak IoT Platform

7.Power Supply

1.Raspberry Pi 4 Model B

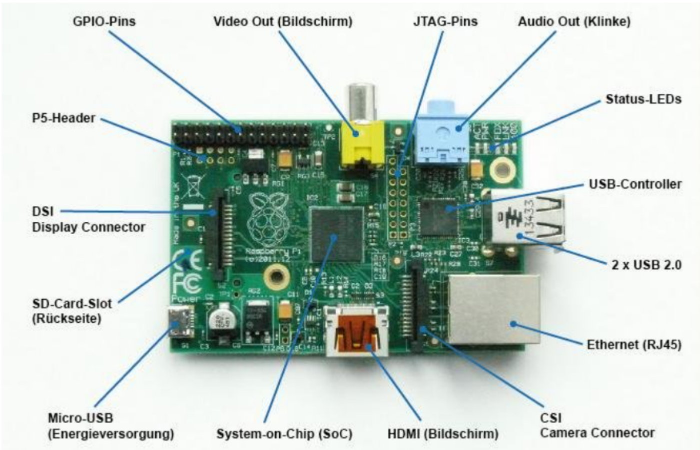

Figure 3.1: Raspberry Pi 4 Model B

1. Description:
The Raspberry Pi 4 Model B (2GB) is a low-cost, credit card-sized computer developed by the Raspberry Pi Foundation. It is designed for education, development, and embedded computing projects and provides a full desktop experience with improved performance over its predecessors.
2. RAM (Memory):
This model comes with 2GB LPDDR4 SDRAM, which allows for smoother performance in multitasking, web browsing, and running light applications, suitable for most educational and basic project needs.
3. Storage:
There is no built-in storage; instead, it uses a microSD card slot for loading the operating system and data storage. It supports high-speed UHS-I cards for better performance.
4. Support for the operating system:
The board supports a variety of OS options like Raspberry Pi OS, Ubuntu, and other Linux-based systems. It can also run lightweight versions of Windows 10 and media centers like LibreELEC.

2.DHT11 Sensor

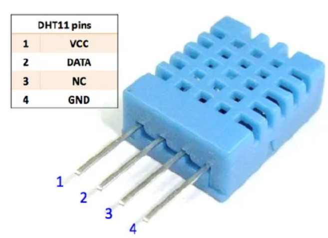

Figure 3.2: DHT11 Sensor

1. Description:
The DHT11 is a basic, ultra-low-cost digital sensor used for measuring temperature and humidity. It is widely used in weather monitoring and home automation systems due to its simplicity and reliability.

2. Function:

The sensor detects temperature and relative humidity from the surrounding air and sends the data digitally to a microcontroller or Raspberry Pi. It uses a capacitive humidity sensor and a thermistor for sensing, providing digital output without the need for analog-to-digital conversion.

3. Temperature Range:

It can measure temperature from 0°C to 50°C with an accuracy of ±2°C. This range is sufficient for most indoor and general environmental monitoring applications.

4. Humidity range:

The DHT11 can measure relative humidity from 20 to 90 Percentage with ±5 Percentage accuracy. It is suitable for detecting moisture levels in the air for comfort or plant/environment monitoring.

3.Raindrop Sensor

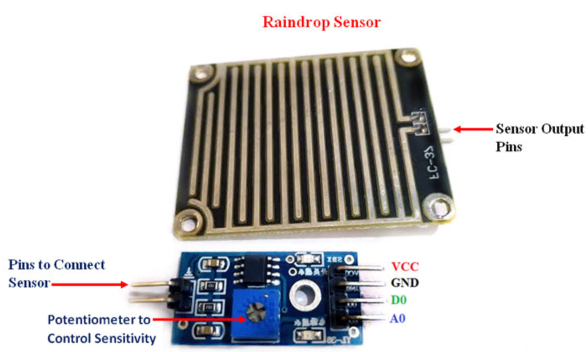

Figure 3.3: Raindrop Sensor

1. Description:

The Raindrop Sensor is a simple weather sensor used to detect the presence and intensity of rainfall. It is commonly used in IoT-based weather stations, irrigation systems, and smart roofs.

2. Function:

This sensor detects water droplets falling on its metal-coated surface and outputs an analog or digital signal based on the presence or absence of rain. It helps trigger actions like closing windows or pausing irrigation when rain is detected.

3. Working Principle:

It works on the principle of electrical conductivity. When raindrops fall on the conductive surface (the sensing plate), the resistance between the tracks changes, allowing it to detect rain intensity.

4. Interface Pins:

The module generally has 4 pins: VCC, GND, Analog Output (AO), and Digital Output (DO). These pins allow easy connection with microcontrollers for reading sensor values.

4.16*2 LCD Display

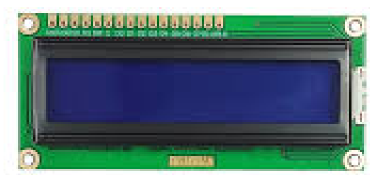

Figure 3.4: 16*2 LCD Display

1. Description:

The 16×2 LCD Display is a widely used alphanumeric screen that can display 16 characters per line on 2 lines. It is based on the HD44780 controller and is popular in embedded and IoT systems for displaying text output.

2. Function:

Its main function is to visually display data such as sensor values, messages, or system status. It receives characters or commands from a microcontroller and shows them in a readable format on its screen.

3. Display Type:

It is a monochrome, dot-matrix display that uses a backlit liquid crystal display.

The text is formed by a 5×8 dot arrangement for each character.

4. Characters Displayed:

It can show a total of 32 characters (16 per row × 2 rows). Each character space consists of a 5x8 dot matrix, and custom characters can also be created using CGRAM.

5. I2C

Figure 3.5: I2C Module

1. Description:

I²C (Inter-Integrated Circuit) is a two-wire serial communication protocol used for short-distance communication between ICs and microcontrollers. It was developed by Philips (now NXP) and is widely used in embedded systems.

2. Function:

The main function of I²C is to enable multiple devices (sensors, displays, EEPROMs, etc.) to communicate with a microcontroller over just two wires: SDA (data) and SCL (clock). It supports master-slave communication.

3. Device Addressing:

Each device on the I²C bus has a unique address (usually 7-bit or 10-bit), allowing multiple devices (typically up to 127) to coexist on the same bus.

4. Speed:

Standard I²C speeds are 100 kbps (Standard Mode), 400 kbps (Fast Mode), 1 Mbps (Fast Mode Plus), and up to 3.4 Mbps (High-Speed Mode), depending on the devices used.

6.ThingSpeak IoT Platform

1. Description:

ThingSpeak is an open-source IoT analytics platform and web service developed by MathWorks. It allows devices to store, analyze, and visualize sensor data in real-time using the cloud.

2. Function:

The primary function of ThingSpeak is to collect data from IoT devices over the internet and provide visualization, alerts, and basic analytics through customizable dashboards and MATLAB integration.

3. Cloud-Based Platform:

Being a cloud-based platform, ThingSpeak enables remote monitoring and control of IoT systems from anywhere via a web browser or mobile device using internet connectivity.

4. Real-Time Data Updates:

ThingSpeak supports real-time data logging with update intervals as low as 15 seconds (for free accounts), allowing near-live monitoring of sensor values.

7.Power Supply

Figure 3.6: Power Supply

1. Description:

A lithium battery is a type of rechargeable or non-rechargeable battery that uses lithium as its primary component. It is widely used in portable electronics, IoT devices, and electric vehicles due to its high energy density.

2. Function:

The primary function of a lithium battery is to store electrical energy and release it when needed to power electronic circuits, sensors, and devices. It offers stable voltage output and long operating times.

3. Chemistry Type:

Lithium batteries come in various chemistries such as Lithium-ion (Li-ion), Lithium Polymer (Li-Po), and Lithium Iron Phosphate (LiFePO4), each with different energy densities and safety profiles.

4. Voltage Range:

A typical lithium-ion cell has a nominal voltage of 3.7V and can go up to 4.2V when fully charged. Lithium primary (non-rechargeable) cells usually offer 3.0V.

Chapter 4

Circuit Diagram

4.1 Circuit Diagram

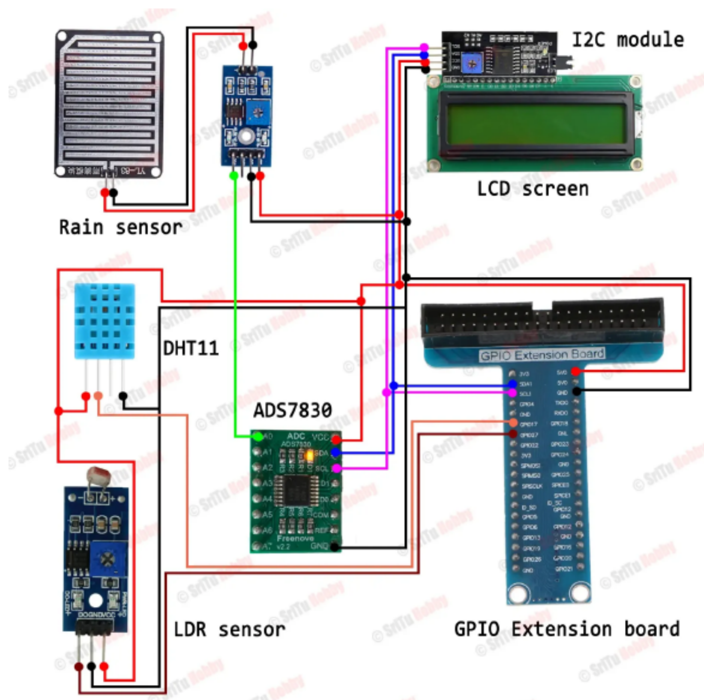

Figure 4.1: Circuit Diagram

This circuit diagram showcases a weather monitoring system based on a Raspberry Pi, utilizing various sensors to gather real-time environmental data. The system reads temperature, humidity, light intensity, and rainfall information and displays the data on an LCD screen. It can also be extended to send the data to the cloud using IoT protocols. Here's a detailed explanation of each component and how they are connected in the circuit:

1. Raspberry Pi with GPIO Extension Board At the heart of the system is the Raspberry Pi, a powerful microcontroller capable of running Python scripts and interfacing with digital and analog components. To simplify connections, a GPIO Extension Board is used. This board extends the GPIO pins into a more accessible and organized layout, helping in cleaner and more manageable wiring.

   - Purpose: Acts as the main controller.
   - Power Supply: 5V and GND pins are used to power external modules and sensors.

2. DHT11 – Temperature and Humidity Sensor The DHT11 sensor is used for measuring ambient temperature and humidity.

   Connections:

   - VCC → 3.3V on GPIO Extension.
   - GND → GND on GPIO Extension.
   - DATA Pin → Connected to a GPIO pin (via a red wire) for data reading.
   - Output: Digital signal representing both temperature and humidity.

   This sensor periodically sends data to the Raspberry Pi, where it is processed and displayed.

3. Rain Sensor Module The rain sensor consists of a rain detection board and a signal conditioning module. It detects the presence and intensity of rainfall.

   Connections:

   - VCC → 5V from GPIO Extension.
   - GND → GND.
   - AO/DO (Analog or Digital Output) → Connected to ADS7830 for analog reading (if using AO).

   It provides analog signals that vary depending on water droplets on the surface, enabling intensity monitoring.

4. LDR Sensor – Light Dependent Resistor The LDR sensor detects ambient light levels and is often used to infer time of day (e.g., daylight vs. nighttime).

   Connections:

   - VCC → 5V.
   - GND → GND.
   - Analog Output → Connected to ADS7830 (an analog-to-digital converter).

   As Raspberry Pi lacks analog input capability, this sensor's output is routed through an ADC module.

5. ADS7830 – Analog to Digital Converter The ADS7830 is an 8-bit ADC module that enables the Raspberry Pi to read analog signals from sensors such as the rain sensor and LDR.

   Connections:

   - VCC → 3.3V.
   - GND → GND.
   - SCL and SDA → Connected to the corresponding I2C pins on the GPIO Extension.
   - AINx pins → Connected to analog output of LDR and Rain sensor.

   The ADC communicates with the Raspberry Pi via the I2C protocol, converting analog voltage levels into readable digital values.

6. I2C LCD Display (16x2 with I2C Module) The LCD screen is used to display real-time data such as temperature, humidity, light intensity, and rain level. It uses an I2C module, which simplifies wiring by reducing the number of control pins.

   Connections:

   - VCC → 5V.
   - GND → GND.
   - SCL and SDA → Shared with the ADS7830 on the I2C bus.

   This screen provides a visual output of sensor readings, ensuring easy real-time monitoring without needing an external monitor or serial interface.

7. Power Supply Consideration The Raspberry Pi is powered via its micro-USB/USB-C power port, and 5V and 3.3V pins on the GPIO extension are used to power external modules. It is crucial to ensure the sensors are compatible with these voltage levels to avoid damage.

Chapter 5

Algorithm

1. Step 1: Initialization of the system.
2. Step 2: Read temperature and humidity from DHT11/DHT22 sensor.
3. Step 3: Filter out incorrect or fluctuating sensor readings.
4. Step 4: Data Transmission over IoT.
5. Step 5: Display real-time data on an LCD screen.
6. Step 6: Compare sensor values with predefined thresholds.
7. Step 7: Repeat steps 2-6 Process.

Chapter 6

Flowchart

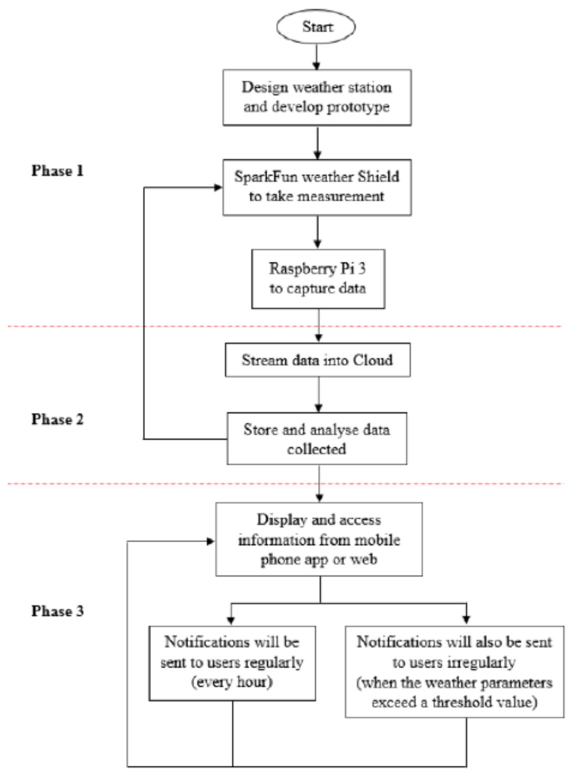

Figure 6.1: Flowchart

Start

The project begins with the initialization of the system.

Phase 1 – Hardware Design and Data Collection

1] Design Weather Station and Develop Prototype

The first step is to design and build a weather monitoring station prototype.

It includes selecting appropriate sensors and modules for weather data measurement.

2] SparkFun Weather Shield to Take Measurement

A SparkFun Weather Shield is used to sense weather parameters such as temperature, humidity, pressure, and wind speed.

This shield is interfaced with the microcontroller or Raspberry Pi.

3]Raspberry Pi 3 to Capture Data

Raspberry Pi 3 is used as the main controller to collect real-time data from the Weather Shield.

It processes the sensor input and prepares it for transmission.

Phase 2 – Data Transmission and Cloud Analysis

4] Stream Data into Cloud

The collected data is transmitted over the internet to a cloud platform using IoT protocols like MQTT or HTTP.

This enables real-time data availability for analysis.

5] Store and Analyse Data Collected

Once the data reaches the cloud, it is stored in a database.

Analytical tools or scripts analyze this data to identify patterns or anomalies (e.g., sudden temperature changes).

Phase 3 – Data Access and Alerts

6] Display and Access Information from Mobile Phone App or Web

Users can access the weather data from a mobile application or web dashboard.

The interface shows current weather conditions and trends.

7] Notifications Sent to Users Regularly (Every Hour)

The system sends periodic weather updates (e.g., hourly) to keep users informed.

8] Notifications Sent to Users Irregularly (When Threshold Exceeds)

If any weather parameter exceeds a defined threshold (e.g., very high wind speed or extreme temperature), immediate alerts are sent to users.

Chapter 7

Result

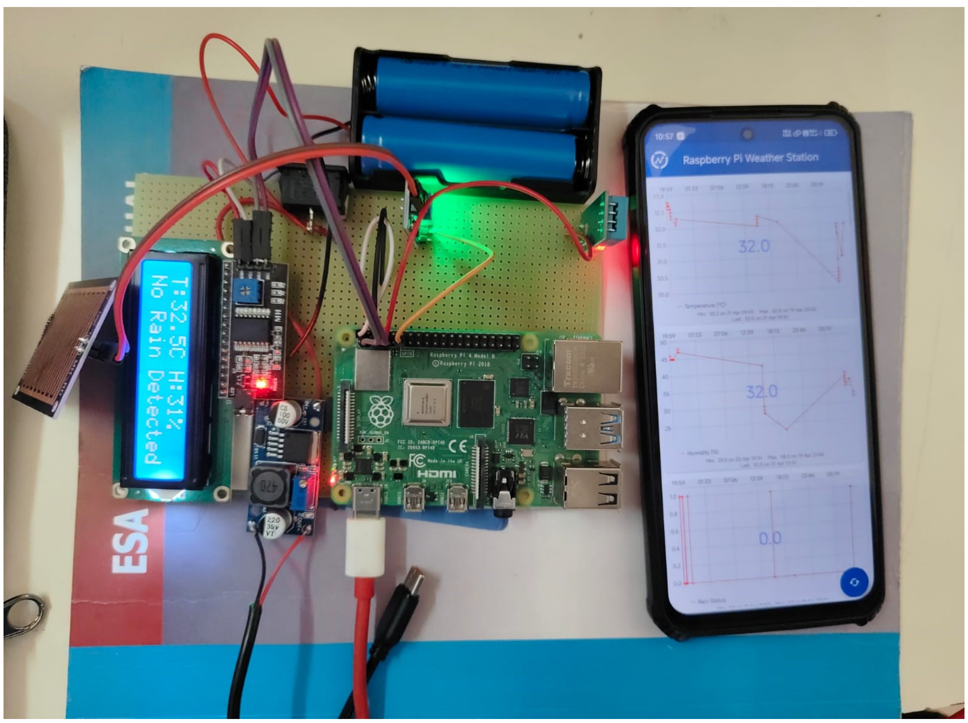

Figure 7.1: Demonstration

The project "Raspberry Pi Based Weather Reporting Over IoT" has been successfully implemented and tested, as evidenced by the setup shown in the image. The system integrates various components, including a Raspberry Pi 4 Model B, temperature and humidity sensors, a rain sensor, an LCD display, and a power source using lithium batteries. The Raspberry Pi serves as the main controller, processing real-time data from the sensors. The measured parameters such as temperature (32.5°C), humidity (31 Percentage), and rain status ("No Rain Detected") are displayed on a 16x2 LCD screen using I2C communication for efficient data transfer.

The data collected by the sensors is transmitted over Wi-Fi to the IoT platform ThingSpeak, which is accessed through a mobile device as shown in the image. The ThingSpeak dashboard presents the weather data in the form of time-series graphs, offering a clear visual representation of environmental trends such as temperature, humidity, and rain rate. This real-time weather monitoring solution not only provides instant data visualization but also enables remote access and analysis through internet connectivity.

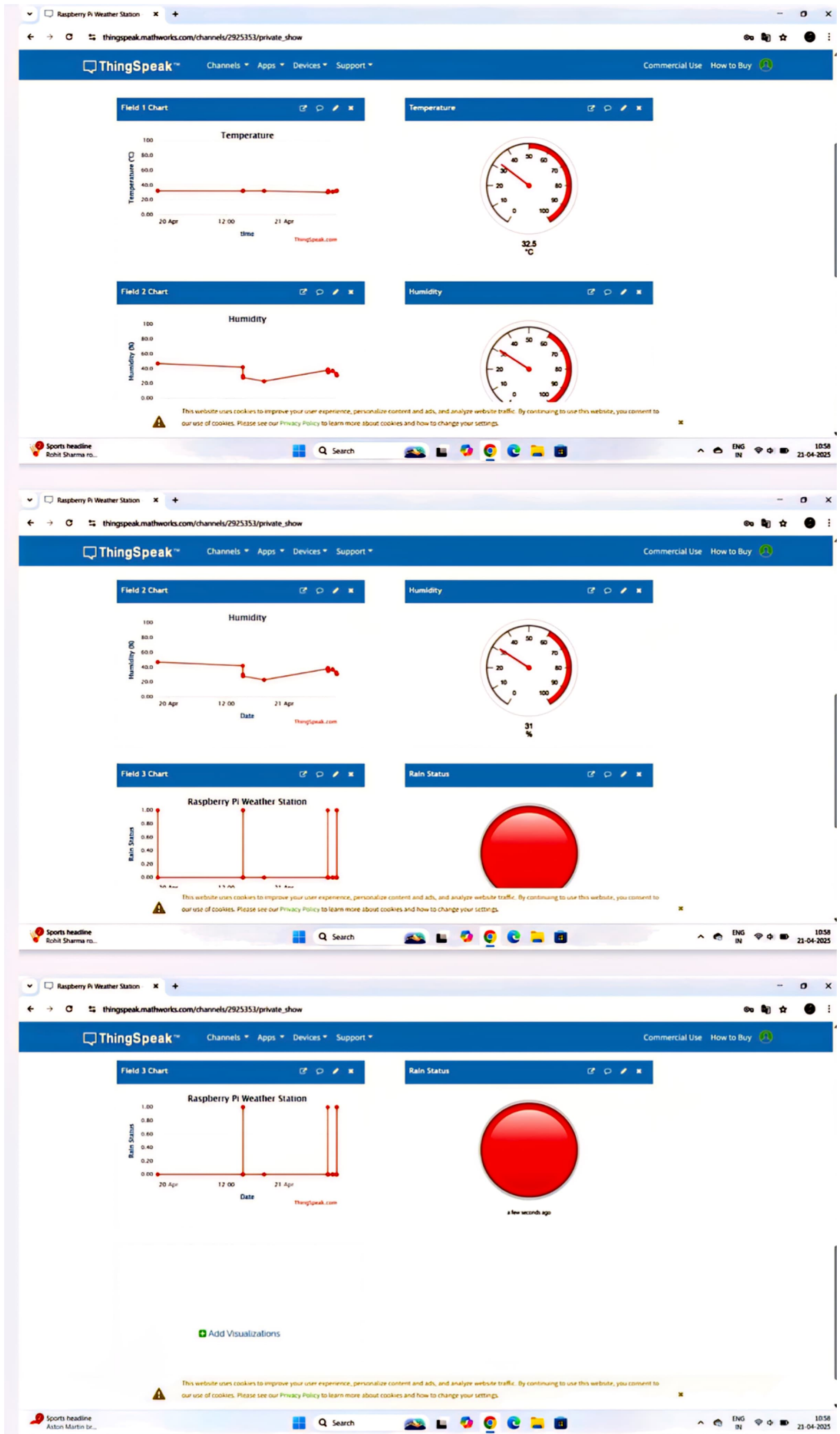

Figure 7.2: Output of the ThingSpeak IoT

Chapter 8

Advantages, Disadvantages and Applications

8.1 Advantages

1.Real-Time Monitoring: Continuously collects and uploads weather data, allowing users to monitor conditions remotely in real time.
2.Low-Cost Solution: Uses affordable hardware like Raspberry Pi and sensors, making it budget-friendly.
3.Portable and Compact: The compact size of the Raspberry Pi makes the system easy to install in various environments.
4.Customizable and Scalable: Can be easily expanded with additional sensors or integrated with other IoT applications.
5.Low Power Consumption: Raspberry Pi operates with minimal power, ideal for remote or solar-powered deployments.
6.Open-Source Platform: Easy to program using Python and supports open-source libraries and tools.

8.2 Disadvantages

1.Limited Processing Power: Raspberry Pi has limited resources compared to full-fledged computers, affecting performance with large datasets or multiple sensors.
2.Internet Dependency: Requires constant internet connection for real-time IoT

data transmission.
3.Environmental Exposure Risk: Outdoor setups require proper casing and weatherproofing to protect electronics.
4.Sensor Accuracy: Low-cost sensors may not offer professional-grade accuracy or long-term durability.
5.Limited Range (Wi-Fi): Wireless connectivity may be limited in rural or remote areas without signal boosters.
6.Data Security: Without encryption or secured servers, data could be vulnerable to breaches or misuse.

8.3 Applications

1.Smart Agriculture:
Helps farmers monitor weather conditions like humidity, temperature, and rainfall to optimize irrigation, planting, and harvesting.
2.Environmental Monitoring:
Deployed in forests, wildlife reserves, or industrial areas to track climate patterns and detect pollution or hazardous conditions.
3.Smart Cities:
Used in urban planning and management to improve public safety, traffic management, and disaster preparedness by monitoring microclimate data.
4.Remote Weather Stations:
Ideal for setting up in remote or hilly regions where human monitoring is not feasible, using solar power and wireless connectivity.
5.Disaster Warning Systems:
Acts as an early-warning mechanism by detecting sudden changes in temperature, pressure, or humidity that indicate extreme weather events.
6.Educational and Research Projects:
Used by students and researchers to study climate data, sensor integration, data logging, and IoT platforms.

Chapter 9

Future Scope

A Raspberry Pi-based weather reporting system over IoT has a future scope encompassing enhanced accuracy, broader applications, and integration with other systems, including agriculture and environmental monitoring, potentially leading to more efficient resource management and disaster preparedness. Here's a more detailed look at the potential future of such a project:

1. Advanced Sensors: Integrate more specialized sensors to measure parameters like soil pH, atmospheric pressure, solar radiation, wind speed and direction, and rainfall.
2. Machine Learning: Incorporate machine learning algorithms to analyze his- torical weather data and make more accurate predictions.
3. Data Visualization: Develop user-friendly dashboards and mobile applications to visualize weather data and receive alerts.
4. Agriculture: Optimized Irrigation: Use soil moisture and rainfall data to determine optimal irrigation schedules, preventing over- or under-watering.
5. Pest Control: Integrate with pest control systems to optimize timing and application of pesticides.
6. Fertilization: Use weather data to optimize fertilizer application, leading to better crop yields. Environmental Monitoring:
7. Air Quality: Integrate sensors to monitor air quality parameters like CO_2, ozone, and particulate matter.
8. Water Quality: Monitor water levels, temperature, and pollutants in rivers and lakes.

9. Disaster Preparedness: Use weather data to predict and prepare for extreme weather events like floods, droughts, and wildfires. Urban Planning:

10. Traffic Management: Use weather data to optimize traffic flow and alert drivers of hazardous conditions.

11. Energy Management: Use weather data to optimize energy consumption in buildings and infrastructure.

Chapter 10

Conclusion

The project ”Raspberry Pi Based Weather Reporting Over IoT” successfully demonstrates a cost-effective, scalable, and real-time solution for monitoring weather conditions using embedded systems and IoT technology. By integrating various environmental sensors with the Raspberry Pi, the system accurately collects key parameters such as temperature, humidity, and atmospheric pressure. This data is transmitted and visualized on an IoT platform like ThingSpeak, enabling users to remotely monitor weather conditions from anywhere with internet access.

The use of Raspberry Pi offers multiple advantages, including low power consumption, compact size, and Python programmability, making it ideal for continuous monitoring applications. The system is particularly useful for applications in agriculture, environmental monitoring, smart cities, and remote weather stations.

Overall, the project highlights the practical potential of combining open-source hardware with IoT platforms to create intelligent, real-time, and automated weather monitoring systems that are easy to deploy and maintain.

Chapter 11

References

1) Bhagat AM, Thakare AG, 1 KAM, Muneshwar NS, Choudhary PV(2019) IOT Based Weather Monitoring and Reporting System Project. International Journal of Trend in Scientific Research and Development(IJTSRD).

2) U. Hariharan and K. Rajkumar, " protract Network Lifetime Using Dynamic Routing for Wireless Sensor Networks, " INTERNATIONAL JOURNAL OF SCIENTIFIC TECHNOLOGY RESEARCH, vol. 9, no. 3, pp. 7166 – 7169, Mar. 2020.

3) Joseph FJJ(2019) IoT Grounded Weather Monitoring System for Effective Analytics. International Journal of Engineering and Advanced Technology(IJEAT).

4) Muck PY, Homam MJ, mariyam@uthmedumy(2018) Iot Grounded Weather Station Using Raspberry Pi 3. International Journal of Engineering Technology.

5) S. S. Sengar, U. Hariharan, and K. Rajkumar, " Multimodal Biometric Authentication System using Deep Learning Method, " 2020 International Conference on Arising Smart Computing and Informatics(ESCI), 2020.

6) Jadhav Gaurav, Jadhav Kunal, Nadlamani Kavita(2016) Environment Monitoring System using Raspberry Pi. International Research Journal of Engineering Technology(IRJET).

7) K. Rajkumar and V. Dhanakoti, "Methodological Survey to Improve the Secure Data Storage in Cloud Computing," 2020 International Conference on Emerging Smart Computing and Informatics (ESCI), 2020.

8) K. Rajkumar and V. Dhanakoti, "Methodological Survey to Improve the Secure Data Storage in Cloud Computing," 2020 International Conference on Emerging

Smart Computing and Informatics (ESCI), 2020.
9) Rana, D., Rajput, R. (2017). "Raspberry Pi Based Weather Monitoring System Using Internet of Things." International Research Journal of Engineering and Technology (IRJET), 4(4), 2215–2219.
10) Monk, S. (2015). Programming the Raspberry Pi: Getting Started with Python (2nd ed.). McGraw-Hill Education.
11) Holler, J., Tsiatsis, V., Mulligan, C., Karnouskos, S., Avesand, S., Boyle, D. (2014). From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence. Academic Press.
12) Raspberry Pi 4 Model B Datasheet – Raspberry Pi Foundation.

Chapter 12

Publication Details

1.Paper Title: "RASPBERRY PI BASED WEATHER REPORTING OVER IOT"
Journal Name: International Research Journal of Modernization in Engineering Technology and Science (IRJMETS)
Volume and Issue: Volume 07, Issue 04
Publication Date: April 2025
Paper ID: IRJMETS70400045037
e-ISSN: 2582-5208
DOI: https://www.doi.org/10.56726/IRJMETS71939
Journal Website: www.irjmets.com
Certificates:

IRJMETS

International Research Journal Of Modernization in Engineering Technology and Science

(Peer-Reviewed, Open Access, Fully Refereed International Journal)

e-ISSN: 2582-5208

Ref: IRJMETS/Certificate/Volume 07/Issue 04/70400045037

DOI: https://www.doi.org/10.56726/IRJMETS71939

Date: 07/04/2025

Certificate of Publication

This is to certify that author "Kashid Vishal" with paper ID "IRJMETS70400045037" has published a paper entitled "RASPBERRY PI BASED WEATHER REPORTING OVER IOT" in International Research Journal Of Modernization In Engineering Technology And Science (IRJMETS), Volume 07, Issue 04, April 2025

IRJMETS Impact Factor 8.187

Editor in Chief

We Wish For Your Better Future

www.irjmets.com

Figure 12.1: Certificate 1

IRJMETS

International Research Journal Of Modernization in Engineering Technology and Science

(Peer-Reviewed, Open Access, Fully Refereed International Journal)

e-ISSN: 2582-5208

Ref: IRJMETS/Certificate/Volume 07/Issue 04/70400045037

DOI: https://www.doi.org/10.56726/IRJMETS71939

Date: 07/04/2025

Certificate of Publication

This is to certify that author "Kate Sagar" with paper ID "IRJMETS70400045037" has published a paper entitled "RASPBERRY PI BASED WEATHER REPORTING OVER IOT" in International Research Journal Of Modernization In Engineering Technology And Science (IRJMETS), Volume 07, Issue 04, April 2025

IRJMETS Impact Factor 8.187

Editor in Chief

We Wish For Your Better Future

www.irjmets.com

Figure 12.2: Certificate 2

IRJMETS

International Research Journal Of Modernization in Engineering Technology and Science

(Peer-Reviewed, Open Access, Fully Refereed International Journal)

e-ISSN: 2582-5208

Ref: IRJMETS/Certificate/Volume 07/Issue 04/70400045037

DOI: https://www.doi.org/10.56726/IRJMETS71939

Date: 07/04/2025

Certificate of Publication

*This is to certify that author **"Lalage Tushar"** with paper ID **"IRJMETS70400045037"** has published a paper entitled "RASPBERRY PI BASED WEATHER REPORTING OVER IOT" in **International Research Journal Of Modernization In Engineering Technology And Science (IRJMETS), Volume 07, Issue 04, April 2025***

Editor in Chief

IRJMETS Impact Factor 8.187

We Wish For Your Better Future

www.irjmets.com

Google scholar issuu Academia.edu doi Crossref Content Registration

Figure 12.3: Certificate 3